The training advisor: a specification of `advisor.py`

Describes `src/training/advisor.py` against exercise dictionary version 9.
Spec: `docs/training/GUIDE.md` Part B. Parameter review: `docs/training/AUDIT.md`.

Contents

1 What this is **1**

2 Axes **2**

3 The exercise dictionary **2**

4 The log and exposures **3**

5 Deficits **3**
5.1 Pattern weights 4

6 Overdue overrides: the guarantee layer **5**

7 Candidate selection **6**

8 Scoring **6**

9 Hard and soft session rules **9**

10 Rep scheme and rep count **9**

11 The `recommend()` verdict **10**

12 A worked example **10**
12.1 A reduced dictionary 11
12.2 A five-event log and the resulting deficits 11
12.3 Normalized deficits and the coupling term 12
12.4 Scoring two cells: overdue versus not 12
12.5 A second pick: resource overlap, now properly scaled 13
12.6 The `recommend()` verdict on this state 14

Needs audit. This entire document describes a retired scoring model. Before any real training log existed, `advisor.py` was rebuilt around a single continuous-decay EWMA per cell (`score = -log(ewma/k)`) and one hard overdue clock, replacing the raw-deficit/window/staleness/ normalization/γ/resource-interference machinery this document formalizes throughout. The resource-cost matrix R and its

interference scoring are gone from the model entirely, not merely re-derived. See `docs/training/GUIDE.md`'s Scoring and Overdue sections for the live model, and `docs/training/STATUS.md` for why the reversal happened before this document was updated to match. Every section below this point — the deficit derivation, the overdue-clock table, the interference formula, the worked example — describes dictionary version 7–9 and is kept only as a correct record of what that model did, not as current documentation. A full rewrite against the live model is tracked but not yet done.

1 What this is

The advisor is a ranking function, not a program generator: it answers "what is most needed today," and a human decides how many pieces to do and designs the actual session. Nothing here is a claim about training science beyond what `docs/training/GUIDE.md` states explicitly — this document only formalizes what the code computes.

Grey boxes throughout mark a value the code needs to run but that `GUIDE.md` either left as a free choice or never specified at all — these are exactly the items in `docs/training/AUDIT.md`, placed here at the point in the math where each one enters. Blue boxes mark a bug that was found (via a two-month closed-loop simulation) and already fixed; they document a correctness constraint the implementation must satisfy, not an open question.

2 Axes

Two small finite sets index everything below.

$$\mathcal{P} = \{\text{hinge}, \text{push}, \text{squat}, \text{pull}, \text{flexion}, \text{brace}, \text{engine}\} \qquad \text{(patterns, 7)}$$

$$\mathcal{S} = \{\text{light}, \text{heavy}, \text{hhr}\} \qquad \text{(styles, 3)}$$

`squat`, `hinge`, `push`, and `pull` name themselves; `flexion` is loaded trunk/hip flexion through range (archetype: toes-to-bar, V-up) and `brace` is resisting movement under load (planks, loaded carries, the Turkish get-up). The last two were a single `midline` pattern until dictionary version 8 — see §5.1. `engine` is restored in version 9 in a deliberately narrow form: it contains *only* exercises whose E row would otherwise be empty (running, biking, rowing, double-unders, burpees). That restriction is what makes it compatible with E's row-stochasticity — since no member carries any other pattern weight, crediting engine never forces a split that would dilute an existing pattern. A general engine pattern would: giving a thruster engine credit means subtracting it from squat and push, because rows sum to at most 1 and weights split volume rather than duplicating it. The first four are Pat Sherwood's CrossFit Linchpin gross movement patterns (originally coded `BP`/`PFG`/`PR`/`UBP`/`ML`) renamed to match the "squat/hinge/push/pull" framing common in general S&C (Dan John and others). Sherwood's "engine" was originally dropped, on the grounds that it classifies energy system rather than movement and duplicated the modality axis — the latter being the styles axis together with the block's logged time-domain fields. Both objections stand against a *general* engine pattern, and neither survives the narrow one above, which is why running and biking no longer sit in the residual. `hhr` ("heavy at high heart rate") is a stimulus, not a session type: load carried while breathing hard, roughly 60–70% of 1RM for reps inside a workout.

3 The exercise dictionary

Let $\mathcal{E}$ be the set of exercises (63 entries in dictionary version 9). Each exercise $e \in \mathcal{E}$ carries:

- $E_e : \mathcal{P} \to [0,1]$, the **pattern-weight row**, with $\sum_{p \in \mathcal{P}} E_{e,p} \leq 1$. The unallocated mass $1 - \sum_p E_{e,p}$ is residual — volume the model doesn't attribute to any pattern. A squat clean is 0.5 hinge $+ 0.5$ squat; a thruster is 0.5 squat $+ 0.5$ push; a clean & jerk is 0.5 hinge $+$ 0.3 push $+ 0.2$ squat. These three are the only rows `GUIDE.md` gives explicitly; every other row is this project's draft, reviewed separately from the constants in this document (see `docs/training/AUDIT.md`'s closing note).
- $R_e : \mathcal{R} \to \mathbb{R}_{\geq 0}$, the **resource-cost row**, over $\mathcal{R} = \{\text{knee}, \text{post}, \text{grip}, \text{shldr}, \text{lats}, \text{trunk}, \text{hang}\}$ (7 resources — `cns` and `aero` both dropped, `hang` added; see Section 8's reversal note). `aero` went in version 9: it had been authored for exactly four exercises, all monostructural with an empty E row, so it was collinear with "is a monostructural exercise" — a family label rather than a cost. Its removal left running and double-unders with empty R rows, which the $\|R_e\| = 0$ guard in interference silently turns into zero similarity with everything, so real tissue costs were authored in its place. `hang` is capsule/tendon fatigue from hanging and swinging from a bar or rings (kipping pull-ups, toes-to-bar, muscle-ups, rope climbs), kept separate from `shldr` (pressing tension) and `grip` (crushing/holding load) since the injury mechanism differs from either. R_e itself is *not* normalized — it is spend, not membership, so a heavier session costs more in absolute terms; only its use inside interference (below) is normalized.
- $A_e \subseteq \mathcal{S}$, the **admissible styles** — which of light/heavy/hhr the exercise can be used for. This is what generates the cell mask below: a cell nothing can fill never accrues deficit.
- a **laterality** in $\{\text{bilateral}, \text{unilateral}\}$ and an **implement** (barbell, dumbbell, kettlebell, bodyweight, ...), used for candidate selection and logging, not for scoring directly.

Neither E nor R is ever fit from data — they are authored judgment, versioned, held fixed. A stable-but-wrong map preserves relative trends; a map that drifts does not.

4 The log and exposures

Every logged set carries a date, a style, an exercise id, and (for unilateral exercises) a side. An **exposure event** is a distinct $(\text{date}, s, e) \in \text{Date} \times \mathcal{S} \times \mathcal{E}$ triple — multiple sets of the same exercise on the same day (a 5×5) collapse to one event, and a unilateral exercise logged on both sides on the same day also collapses to one event, not two.

> **Found and fixed.** The implementation originally summed $E_{e,p}$ once per logged *set row* rather than once per exposure event, so a 5×5 counted as five exposures instead of one, and a unilateral exercise logged L+R counted double. Fixed by collapsing to the event definition above before applying pattern weights. Two regression tests guard this (`test_multiple_sets_same_day_count_as_one_exposure`, `test_unilateral_sides_same_day_count_as_one_exposure`).

Fix a trailing window of W days ending on the query day t:

$$X_{p,s}(t) \;=\; \sum_{\substack{(\text{date}, s', e)\ \text{exposure event} \\ s' = s,\ \text{date} \in (t-W,\, t]}} E_{e,p}$$

the pattern-weighted count of exposures to pattern p at style s within the window.

> **Needs audit.** `WINDOW_DAYS` = 42. `GUIDE.md` calls this "the most consequential single number" in the whole model, but doesn't second-guess the value itself. Every deficit and staleness figure below is relative to this window, so it is worth confirming it still matches your actual training rhythm before trusting anything downstream.

5 Deficits

Each style has a weekly **target rate** τ_s (exposures per week per pattern), each pattern a dimensionless **pattern weight** ω_p, and the target for cell (p, s) over the window is $\tau_s \, \omega_p \, w$ where $w = W/7$ is the window length in weeks.

s	τ_s (per week per pattern)	source	note
light	1.35	`GUIDE.md`, trust	residual style, targeted highest
heavy	0.50	`GUIDE.md`, trust	goal: heavy once per fortnight
hhr	0.15–0.25	`GUIDE.md`, free choice	this code uses 0.20

5.1 Pattern weights

$$\omega_p = \begin{cases} 0.25 & p = \text{brace} \\ 1 & \text{otherwise.} \end{cases}$$

Until dictionary version 8 targets were keyed by style alone and applied uniformly across patterns, i.e. $\omega_p \equiv 1$. That was harmless while $|\mathcal{P}|$ was fixed, and became load-bearing the moment it wasn't: splitting `midline` into `flexion` and `brace` under $\omega_p \equiv 1$ would have taken total weekly demand from

$$|\mathcal{P}| \sum_s \tau_s = 5 \times 2.05 = 10.25 \quad \text{to} \quad 6 \times 2.05 = 12.3$$

against a supply of roughly 12 pattern-bearing pieces per week — the entire capacity slack, spent silently by a taxonomy change that looks purely descriptive. With $\omega_{\text{brace}} = 0.25$, demand is

$$\sum_p \omega_p \sum_s \tau_s = (5 + 0.25) \times 2.05 = 10.76,$$

and the slack survives — though only until version 9 added engine, which has no heavy cell but does have light and hhr ones, taking demand to 12.31 against the same ≈ 12. The slack is now slightly negative; see the audit box below.

> **Needs audit.** $\omega_{\text{engine}} = 1$ leaves total demand at 12.31 per week against a supply of roughly 12 pattern-bearing pieces — i.e. the model is now **over-subscribed**, and the closed-loop simulation shows it: `push/light` and `squat/light` finish the run *behind* target, the first positive end-state deficits any simulation has produced. Lower ω_{engine}, raise the supply estimate on the argument that a run does not compete for a lifting slot, or accept over-subscription deliberately.

Needs audit. $\omega_{\text{brace}} = 0.25$ is a **free choice with nothing behind it but the arithmetic above** — it was chosen to fit the capacity budget, not derived from a stated goal about how much carrying and bracing is wanted. It is also, as of this writing, *not what the model actually delivers*: the 7-day any-exposure clock imposes a floor of roughly one exposure per week on every pattern, which outvotes ω_{brace} entirely. See the audit box in Section 6.

Needs audit. `TARGET_RATE["hhr"]` = 0.20. `GUIDE.md` gives a range (0.15–0.25, "free choice; 0.25 ≈ 2:1 heavy:hhr") and leaves the exact value to you. This is the midpoint, not a derived number — pick your own ratio.

A cell (p, s) is **masked** (excluded from everything below) unless some exercise can fill it:

$$\text{admissible}(p, s) \iff \exists\, e \in \mathcal{E} : E_{e,p} > 0 \text{ and } s \in A_e.$$

For admissible cells, the deficit is

$$D_{p,s}(t) = \tau_s\, \omega_p\, w \;-\; \widetilde{X}_{p,s}(t), \qquad \widetilde{X}_{p,s}(t) = \begin{cases} X_{p,\text{heavy}}(t) + \eta \cdot X_{p,\text{hhr}}(t) & s = \text{heavy} \\ X_{p,s}(t) & \text{otherwise,} \end{cases}$$

where η is the **hhr credit**: hhr work partially, but never fully, substitutes for fresh heavy work toward the heavy cell's own deficit — hhr's own cell is unaffected by η and is scored against its own target independently.

Needs audit. η = `HHR_CREDIT` = 0.5. `GUIDE.md` gives both the value and the physiological reasoning explicitly (loaded conditioning at 60–70% under fatigue trains work capacity and strength-endurance but contributes little to maximal strength, which needs proximity to max plus a rested nervous system), so this is probably right as specified. Flagged anyway because `GUIDE.md` itself says "if the priority ever shifts toward maximal strength, lower this credit" — that call belongs to you, not the code.

Targets *rank*; they are not reachable goals to report distance from. With demand near capacity some cells always show shortfall by design, and that is fine — only the ordering needs to be right.

6 Overdue overrides: the guarantee layer

Deficits rank; they do not enforce a hard floor on their own. A separate guarantee layer gives a cell past a frequency clock a bonus large enough to outrank everything else in the score:

clock	threshold	satisfied by	applies to cell (p, s) when
any exposure	7 days	any style	always
loaded	14 days	heavy or hhr	$s \in \{\text{heavy}, \text{hhr}\}$, $p \notin \mathcal{X}$
classic heavy	24 days	heavy only	$s = \text{heavy}$, $p \notin \mathcal{X}$

Formally, with $\text{last}(p, \mathcal{S}')$ the most recent exposure date to pattern p at any style in $\mathcal{S}' \subseteq \mathcal{S}$ (or $-\infty$ if never):

$$\text{overdue}(p, s, t) = \begin{cases} B & t - \text{last}(p, \mathcal{S}) > 7 \\ B & p \notin \mathcal{X},\ s \in \{\text{heavy}, \text{hhr}\},\ t - \text{last}(p, \{\text{heavy}, \text{hhr}\}) > 14 \\ B & p \notin \mathcal{X},\ s = \text{heavy},\ t - \text{last}(p, \{\text{heavy}\}) > 24 \\ 0 & \text{otherwise,} \end{cases}$$

checked in that order (the first satisfied condition wins), where $\mathcal{X} = \{\text{flexion}, \text{brace}, \text{engine}\}$ is the set of **clock-exempt** patterns. Both are exempt from the 14- and 24-day clocks — every other pattern loads the trunk incidentally, so a dedicated heavy-trunk target would overstate the need, and `brace` additionally consists of movements that are loaded by nature and would otherwise trip the classic-heavy clock unprompted — but *not* from the 7-day any-exposure clock.

> **Needs audit.** That last exemption boundary is currently doing something unintended. The 7-day clock implies a floor of roughly one exposure per week, while $\omega_{\text{brace}} = 0.25$ asks for about half that. The clock always wins, so `brace`'s effective rate is about twice its stated target. Either $\mathcal{X}$'s members need a longer any-exposure threshold, or ω_{brace} should rise to match a weekly floor. See `docs/training/AUDIT.md`.

The loaded and classic-heavy clocks only bonus the styles that can actually resolve them: a light pick can't clear a "loaded" debt, so $\text{overdue}(p, \text{light}, t)$ never receives the 14- or 24-day bonus, only the 7-day one.

> **Needs audit.** B = `OVERDUE_BONUS` = 1000.0. `GUIDE.md` specifies only that the bonus must be "large enough to outrank everything"; this is a sentinel constant, not a bound derived from the other terms' ranges. Safe as long as no plausible combination of the terms in Section 8 can approach it, but worth re-checking if the λ's or staleness caps below are ever changed by a large factor.

7 Candidate selection

For an admissible cell (p, s), the eligible exercises are $\{e \in \mathcal{E} : E_{e,p} > 0,\ s \in A_e\}$. The candidate offered for that cell is the **stalest eligible variant**:

$$\hat{e}(p, s, t) = \underset{e \text{ eligible}}{\arg\max} \Big(t - \text{lastlogged}(e),\ E_{e,p},\ e \Big)$$

lexicographically — staleness first (never-logged treated as $+\infty$), pattern weight as tie-break, exercise id as final tie-break for determinism. Because each cell contributes only its stalest variant, the second-best option for that cell is never shown — a known, documented limitation, not a bug.

8 Scoring

For each admissible cell, with $\hat{e} = \hat{e}(p, s, t)$ as the candidate and C the set of exercises already chosen this session:

$$\begin{aligned}
\text{score}(p, s) \;=\; & \widehat{D}_{p,s}(t) \\
& + \; \gamma_p \cdot M_p(t) \\
& + \; \lambda_{\text{cell}} \cdot \frac{\min\big(t - \text{last}(p, \{s\}),\ 21\big)}{21} \\
& + \; \lambda_{\text{uni}} \cdot \frac{\min\big(\max(u_p, 0),\ 3.0\big)}{3.0} \\
& + \; \text{overdue}(p, s, t) \\
& - \; 0.25 \sum_{c \in C} \widehat{R}_{\hat{e}}\, W\, \widehat{R}_c^{\top} \\
& - \; 1.1 \sum_{c \in C} \mathbb{1}\big[\exists\, p' : E_{\hat{e},p'} > 0 \wedge E_{c,p'} > 0\big]
\end{aligned}$$

where $\widehat{R}_e = R_e / \|R_e\|_2$ is e's resource-cost row unit-normalized. with the following pieces.

Reversal: normalized deficit and staleness, λ_{move} removed. $\widehat{D}_{p,s}(t) = D_{p,s}(t)/(\tau_s \omega_p w)$ — the deficit expressed as a fraction of that *cell's* own target. Since exposure is never negative, $D_{p,s} \le \tau_s \omega_p w$ always, so $\widehat{D}$ is bounded above by 1 with no new machinery. Dividing by the pattern-scaled target (rather than $\tau_s w$) extends the same guarantee across patterns now that $\omega_p \not\equiv 1$: a fully-neglected `brace` cell saturates at 1 exactly like a fully-neglected `squat` cell, despite an absolute target a quarter the size. Without it `brace` could never outrank `squat` no matter how long it went untouched — the identical failure this normalization was introduced to fix across styles. This matters because targets vary a lot in absolute size (hhr's is 1.2 over the window, light's is 8.1): comparing raw deficits structurally favours whichever style has the biggest target, exactly the problem `recommend()` (Section 11) already had to correct for one level up. Normalizing fixes it here too: a fully-neglected hhr cell and a fully-neglected light cell both saturate at $\widehat{D} = 1$, on equal footing. Cell staleness and the unilateral shortfall are divided by their own caps (21 days, 3.0) the same way, so every additive term above lives on a comparable $[0, 1]$-ish scale, and the λ's are weights on that scale rather than magnitude-matched placeholders guessed against a moving target.

λ_{move} (movement-level staleness) has been removed entirely. It never affected *which* exercise got offered for a cell — that was always $\hat{e}(p, s, t)$ from Section 7, independent of scoring. Its only job was a secondary nudge to a cell's ranking when its offered candidate happened to be stale even though the cell overall wasn't; GUIDE.md records that an ablation on the discarded prototype had already found it moved outcomes by ~ 0.05 on one seed — removing it is a real simplification, not a loss of a load-bearing mechanism.

Deliberately not done: no unbounded/accumulating staleness term was added to compensate. An accumulating deficit (never capped, only reset by servicing, à la Deficit Round Robin scheduling) would give the same eventual-guarantee property the overdue clocks below provide, but this model does greedy top-1 selection rather than round-robin service, so an unbounded backlog could monopolize the schedule for an extended stretch once large enough. The clocks stay a separate, bounded, fixed- magnitude override rather than a growing one.

Needs audit. Correction from an earlier version of this document: $\gamma_{\text{behind}} = 0.35/\gamma_{\text{over}} = 0.10$ and the resource-overlap coefficient 0.25 below were previously described as trusted GUIDE.md values, reasoning that GUIDE.md states their numbers directly. That conflated "has a specific number attached" with "derived from stated goals." GUIDE.md's own parameter classification places *both* in the tuned-against-synthetic-history, near-noise class — same as the λ's. The asymmetry in γ (behind pulls harder than over-covered pushes back) is the load-bearing idea; the specific magnitudes came from the same ablation process as everything else in this box, not from GUIDE.md's stated requirements.

$\lambda_{\text{cell}} = 0.25$ (was 0.10), $\lambda_{\text{uni}} = 0.10$ (was 0.30) — rescaled for the normalization above, by preserving each term's old ratio to deficit's old maximum: λ_{cell} contributed up to $0.10 \times 21 = 2.1$ against a max deficit of 8.1, a ratio of ≈ 0.26; λ_{uni} contributed up to $0.30 \times 3.0 = 0.9$ against the same 8.1, a ratio of ≈ 0.11. Rounded to clean numbers rather than false precision. Still a guess, just a rescaled one — GUIDE.md's own posture here is "leave conservative until real logs exist."

Newly discovered, not yet resolved: the resource-overlap coefficient (0.25, below) and the same-pattern-repeat coefficient (1.1) were left at their original values, since neither was part of what this normalization pass touched. But both multiply a *raw*, unnormalized magnitude ($RWR^\top$, and a 0/1 indicator), the same kind of quantity λ_{cell} used to be calibrated against before it was rescaled. With deficit now capped at 1 instead of running up to 8, these two penalty terms are relatively far more powerful than before, and can now push a candidate's score negative on their own (the worked example in Section 12 hits exactly this). Whether that's a problem or a reasonable consequence of "resource conflicts should genuinely veto a pick" is an open question, not yet decided.

Coupling across styles. $M_p(t)$ is now the *mean* of $\widehat{D}_{p,s'}(t)$ over p's admissible styles s' (not the sum, as it was before normalization) — keeping it on the same $[-\infty, 1]$-ish scale as $\widehat{D}$ itself:

$$M_p(t) = \frac{1}{|\{s' : \text{admissible}(p, s')\}|} \sum_{s' \in \mathcal{S}} \widehat{D}_{p,s'}(t), \qquad \gamma_p = \begin{cases} 0.35 & M_p(t) > 0 \\ 0.10 & \text{otherwise.} \end{cases}$$

The asymmetry matters: without it, over-coverage in one style starved patterns that had few admissible cells in another.

Unilateral shortfall. $u_p = \texttt{UNILATERAL_TARGET} \cdot w - U_p(t)$, where $\texttt{UNILATERAL_TARGET} = 0.30$ per week per pattern (`GUIDE.md`, trust) and $U_p(t)$ is the windowed, pattern-weighted count of exposure events from unilateral exercises, clamped to $[0, 3.0]$ before normalizing — uncapped, every pattern went maximally behind at once the first time this was implemented, and the whole candidate list went unilateral.

Reversal: resource interference is now cosine similarity. $W \in \mathbb{R}^{8\times 8}$ is the resource-interaction matrix, so $\widehat{R}_{\hat{e}}\, W\, \widehat{R}_c^\top$ — with $W = I$, this reduces to the ordinary cosine similarity between $\hat{e}$ and c's resource-cost profiles, bounded to $[0, 1]$ since no resource cost is negative. This is what lets the model know a bike ride blocks a heavy squat despite sharing no movement pattern.

Two changes from an earlier version of this document, made together. First, `cns` was dropped from $\mathcal{R}$ entirely: it was present in every one of the 64 exercises, making it the single most universally-shared resource and the dominant driver of interference between almost any two picks, regardless of whether they actually competed for the same tissue — also the least concretely groundable of the eight numbers being authored per exercise to begin with. Second, interference used to be the *raw* dot product $R_{\hat{e}}\, W\, R_c^\top$, unbounded and scaling with both vectors' absolute magnitude; once deficit was normalized to $[0, 1]$, this became disproportionately powerful, as Section 12 showed directly (a candidate's score flipped negative purely from resource overlap). Unit-normalizing R before the dot product — i.e. cosine similarity — fixes that by construction, and also changes what the term *means*: from "how much absolute stress do these share" to "how similar are their stress profiles," which is arguably the more correct notion of interference independent of the rescaling motivation.

Needs audit. `GUIDE.md` defines interference as $R\,W\,R^\top$ but never gives W's values. It currently defaults to the 8×8 identity matrix — no assumed cross-resource coupling; `grip` fatigue doesn't spill into `shldr` fatigue, `hang` doesn't spill into either, and so on. That is the most conservative possible choice, not a reasoned one. `hang` is the resource most likely to actually deserve coupling to `shldr`/`grip` (a kipping shoulder is a real overuse pattern, not an independent one) — worth a look once real logs exist. If some resources should interact, edit the matrix directly in `exercises.json`. Separately, `RESOURCE_OVERLAP_COEF` $= 0.25$ is unchanged in value but not in meaning: it now weights a bounded $[0, 1]$ quantity, so it reads as "a fully-overlapping resource profile costs as much as maximal cell staleness" ($= \lambda_{\text{cell}}$) rather than a fraction of an unbounded raw magnitude. Still a guess pending real logs.

Same-pattern repeat. The indicator sum counts how many already-chosen exercises this session share at least one pattern with the candidate. This is a soft penalty, not a mask — back squat then front squat stays reachable if below-parallel is genuinely starved — with coefficient 1.1, of unclear provenance: GUIDE.md's near-noise list doesn't name it explicitly, but it isn't in the trust-these requirements table either.

9 Hard and soft session rules

One rule is a hard mask rather than a scoring penalty: a pattern loaded today (by any already-chosen exercise, at any style) cannot reappear at `hhr` in the same session — no heavy squat followed by squatting at high heart rate. Formally, when generating candidates for $s = \text{hhr}$, the cell (p, hhr) is dropped entirely (not merely penalized) whenever $p \in \bigcup_{c \in C} \{p' : E_{c,p'} > 0\}$.

Every other interaction between already-chosen exercises and a new candidate — same pattern at heavy, shared resource cost — is a soft penalty already folded into the score in Section 8, not a second masking layer.

10 Rep scheme and rep count

Rep schemes apply to `heavy` only (an `hhr` piece gets a rep target, not a scheme). For a pattern p due for heavy work, the suggested scheme is whichever entry in a fixed menu was least recently

used for that pattern at `heavy` — ranked per pattern, not globally, so heavy cleans avoid schemes recently used by heavy deadlifts.

> **Needs audit.** The scheme menu (`REP_SCHEMES`: `5x5`, `3x3x3x3x3`, `5-3-1`, `8x3`, `3RM build`, `10x2`, `4x6`) doesn't come from `GUIDE.md` — it describes schemes being ranked per pattern by staleness but never names an actual menu. This list is a generic placeholder; swap in whatever you actually program.

Rep counts are not prescribed at all: `suggest_rep_count` returns the rolling median of reps logged for a given (exercise, style) pair, or `None` if there is no history yet. Rep counts are learned from the log, never authored, because rep counts are skewed and a median is the right summary; schemes are categorical and cannot be averaged the same way, which is why they are ranked instead.

11 The `recommend()` verdict

`recommend()` answers which *mode* (not cell) is most needed, comparing styles **relative to each style's own target** — light is targeted at nearly three times heavy's rate, so comparing raw deficits would always favor light regardless of actual need. For each style with at least one admissible cell:

$$\rho_s(t) = 100 \cdot \frac{\sum_p D_{p,s}(t)}{\sum_{p:\text{admissible}(p,s)} \tau_s \, \omega_p \, w}$$

the aggregate deficit as a percentage of that style's own aggregate target. Let $s^\star = \arg\max_s \rho_s(t)$.

> **Found and fixed.** Ranking $s^\star$ purely by $\rho_{s^\star}(t) = \max_s \rho_s(t)$ without checking its sign let the verdict declare a style "needed" when *every* style was already over its target. A two-month closed-loop simulation showed exactly this: once every $\rho_s(t)$ went negative (all styles over-covered — an artifact of the simulation forcing a fixed number of pieces per session with no stopping rule), light kept "winning" the comparison every week, with the reported margin *growing*, purely because heavy and hhr were becoming more over-covered *faster* — not because light was falling behind. Fixed: if $\rho_{s^\star}(t) \le 0$, the verdict is unconditionally `no strong signal` with no style named, regardless of the margin to the runner-up. Guarded by `test_recommend_does_not_favour_the_least_overshot_style`.

When $\rho_{s^\star}(t) > 0$, let $\rho_{(2)}(t)$ be the second-highest value and $\mu(t) = \rho_{s^\star}(t) - \rho_{(2)}(t)$ the margin. The verdict bands are

$$\text{band}(t) = \begin{cases} \texttt{clear} & \mu(t) > 25 \\ \texttt{leaning} & \mu(t) > 10 \\ \texttt{no strong signal} & \text{otherwise,} \end{cases}$$

with the style named only in the first two bands — `GUIDE.md` is explicit that most days genuinely have no right answer, and the output should say so rather than force a pick. All three band thresholds (25, 10) are given directly by `GUIDE.md`, not tuned here.

12 A worked example

Everything above is abstract; this section runs the actual pipeline on a small, hand-picked slice of the dictionary and a short synthetic log, so every number below can be checked by hand. This is illustrative only — a reduced 5-exercise dictionary, not the full dictionary itself, chosen to touch every mechanism in the model with round numbers.

Needs audit. This section is still on the pre-version-8 five-pattern set and has not been reworked. Everything above it reflects dictionary version 8 (six patterns, ω_p, $\mathcal{X}$); the arithmetic below was hand-checked against version 7 and is left intact rather than half-migrated, because the two are not reconcilable by renaming: the example's fifth exercise is a kettlebell swing carrying midline = 0.2, and version 8 removed exactly that credit on the grounds that a swing's reactive bracing is a resource cost, not adaptation-specific transfer. Reworking it means choosing a new fifth exercise and recomputing the deficit table, the normalized deficits and both scored cells — worth doing deliberately, not as a search-replace. Read this section as a correct illustration of the *mechanisms*, with stale pattern names.

12.1 A reduced dictionary

Five exercises, one per pattern (kettlebell swing also touches midline as a secondary pattern, showing what a nonzero residual looks like):

exercise	hinge	push	squat	pull	midline	residual	admissible styles
back squat	–	–	1.0	–	–	0.0	heavy, hhr
kettlebell swing	0.4	–	–	–	0.2	0.4	light, hhr
strict pull-up	–	–	–	1.0	–	0.0	light, heavy, hhr
strict press	–	1.0	–	–	–	0.0	heavy, hhr
deadlift	1.0	–	–	–	–	0.0	light, heavy, hhr

this is the matrix E — rows are exercises, columns are patterns, "–" is 0

exercise	knee	post	grip	shldr	lats	trunk	hang
back squat	3	1	–	–	–	2	–
kettlebell swing	–	1	1	–	–	1	–
strict pull-up	–	–	1	–	2	–	1
strict press	–	–	–	3	–	1	–
deadlift	–	3	2	–	–	2	–

this is the matrix R (`aero` omitted — none of these five touch it; `cns` is gone from the model entirely, see Section 8. Strict pull-up's `hang`= 1 doesn't feed into any calculation shown below — none of the interference examples involve it.)

All five are bilateral, so the unilateral-shortfall term below comes out identical across patterns in this example — dictionary version 7 has unilateral variants for all of these; none are included here to keep the arithmetic small. W, the resource-interaction matrix, is the 8×8 identity throughout

(Section 8's audit box) — with $W = I$, $\widehat{R}_a \, W \, \widehat{R}_b^{\top}$ reduces to ordinary cosine similarity, so it is not worth drawing out as a 64-entry table of mostly zeros.

From this dictionary alone, three cells are masked before any log exists at all: **(squat, light)** — no light-admissible squat exercise here — and likewise **(push, light)** and **(midline, heavy)**. (Dictionary version 7 does have light-squat and heavy-midline options; the mask here is an artifact of using only 5 exercises, not a property of the real dictionary.) 12 cells remain admissible.

12.2 A five-event log and the resulting deficits

Take the query day t as "today," with the usual 42-day window ($w = 6$ weeks). The log:

days before t	exercise	style	contributes
3	back squat	heavy	squat += 1.0
5	kettlebell swing	hhr	hinge += 0.4, midline += 0.2
15	strict pull-up	heavy	pull += 1.0
20	deadlift	light	hinge += 1.0
25	strict press	heavy	push += 1.0

All five fall inside the window (the oldest, at day 25, is well clear of the day-42 edge, picked deliberately so none of the near-the-boundary cliff behaviour discussed earlier shows up here and complicates the arithmetic). Windowed exposures $X_{p,s}$, everywhere else 0:

$$X_{\text{squat,heavy}} = 1.0, \qquad X_{\text{hinge,hhr}} = 0.4, \qquad X_{\text{midline,hhr}} = 0.2,$$
$$X_{\text{pull,heavy}} = 1.0, \qquad X_{\text{hinge,light}} = 1.0, \qquad X_{\text{push,heavy}} = 1.0.$$

Targets: $\tau_{\text{light}} \, w = 1.35 \times 6 = 8.1$, $\tau_{\text{heavy}} \, w = 0.50 \times 6 = 3.0$, $\tau_{\text{hhr}} \, w = 0.20 \times 6 = 1.2$. Only hinge's heavy cell picks up the $\eta = 0.5$ hhr credit, from the kettlebell-swing exposure: $\widetilde{X}_{\text{hinge,heavy}} = 0 + 0.5 \times 0.4 = 0.2$; every other heavy cell here has zero hhr exposure to credit, so $\widetilde{X} = X$ for them.

pattern	light	heavy	hhr
hinge	$8.1 - 1.0 = 7.1$	$3.0 - 0.2 = 2.8$	$1.2 - 0.4 = 0.8$
push	*masked*	$3.0 - 1.0 = 2.0$	$1.2 - 0 = 1.2$
squat	*masked*	$3.0 - 1.0 = 2.0$	$1.2 - 0 = 1.2$
pull	$8.1 - 0 = 8.1$	$3.0 - 1.0 = 2.0$	$1.2 - 0 = 1.2$
midline	$8.1 - 0 = 8.1$	*masked*	$1.2 - 0.2 = 1.0$

$D_{p,s}(t)$, the full deficit table for this state

12.3 Normalized deficits and the coupling term

Dividing the table above by each column's own target (8.1/3.0/1.2) gives $\widehat{D}_{p,s}(t)$, and averaging each pattern's admissible entries gives $M_p(t)$ (Section 8):

pattern	light	heavy	hhr	$M_p(t)$
hinge	0.877	0.933	0.667	0.826
push	*masked*	0.667	1.000	0.833
squat	*masked*	0.667	1.000	0.833
pull	1.000	0.667	1.000	0.889
midline	1.000	*masked*	0.833	0.917

Every $M_p(t) > 0$ here, so $\gamma_p = 0.35$ throughout this example.

12.4 Scoring two cells: overdue versus not

Cell (push, heavy). The only pattern-push exercise here, strict press, is also the only eligible candidate, so $\hat{e}$ = strict press regardless of staleness. Pattern push was last touched (any style) 25 days ago, so $t - \text{last}(\text{push}, \mathcal{S}) = 25 > 7$: the 7-day clock alone is already tripped, so overdue = 1000 before the 14- and 24-day clocks are even consulted. With nothing chosen yet this session ($C = \emptyset$, so the overlap and repeat terms are both 0):

$$\begin{aligned}\text{score}(\text{push}, \text{heavy}) &= \underbrace{0.667}_{\widehat{D}} + \underbrace{0.35 \times 0.833}_{\gamma M = 0.292} + \underbrace{0.25 \times \tfrac{\min(25,21)}{21}}_{\lambda_{\text{cell}} = 0.250} \\ &\quad + \underbrace{0.10 \times \tfrac{1.8}{3.0}}_{\lambda_{\text{uni}} = 0.060} + \underbrace{1000}_{\text{overdue}} \\ &= 1001.27.\end{aligned}$$

(The unilateral term is $0.10 \times \min(\max(0.30 \times 6 - 0, 0), 3.0)/3.0 = 0.10 \times 0.6 = 0.060$ for every pattern here, since none of the five exercises are unilateral.)

Cell (hinge, light). Two exercises are eligible — deadlift and kettlebell swing both allow light and both carry hinge weight. Deadlift was last logged 20 days ago; kettlebell swing 5 days ago, *as hhr* — which no longer matters for scoring since λ_{move} is gone, but it still decides which exercise selection offers, and deadlift is stalest. Pattern hinge was touched 5 days ago (that same kettlebell-swing set), so $t - \text{last}(\text{hinge}, \mathcal{S}) = 5 \leq 7$: not overdue — and since this cell's style is `light`, the 14- and 24-day clocks never apply to it regardless of staleness (Section 6):

$$\begin{aligned}\text{score}(\text{hinge}, \text{light}) &= \underbrace{0.877}_{\widehat{D}} + \underbrace{0.35 \times 0.826}_{0.289} + \underbrace{0.25 \times \tfrac{\min(20,21)}{21}}_{0.238} \\ &\quad + \underbrace{0.10 \times \tfrac{1.8}{3.0}}_{0.060} + \underbrace{0}_{\text{overdue}} \\ &= 1.46.\end{aligned}$$

Even though (hinge, light) has the second-largest raw deficit in the whole table, (push, heavy) outranks it by roughly 685× — *more* lopsided than before normalization (it was ∼ 70×), because normalizing shrank the non-overdue scores without touching the overdue bonus at all. Exactly what "a bonus large enough to outrank everything" is supposed to do, and a reminder that $B = 1000$ has even more headroom now (see Section 6's audit box). (Not shown in detail: (hinge, heavy) is *also* overdue in this state, but only via the 24-day classic-heavy clock — hinge has never actually been trained at true heavy style in this log, so $t - \text{last}(\text{hinge}, \{\text{heavy}\}) = \infty$, even though the recent hhr exposure keeps both the 7- and 14-day clocks satisfied. This is the reason the model needs three separate clocks rather than one: hhr work can keep a pattern "warm" without ever satisfying the specific requirement for fresh heavy work.)

12.5 A second pick: resource overlap, now properly scaled

Suppose (push, heavy) is chosen first, so $C = \{\text{strict press}\}$. Now score (squat, hhr) — candidate: back squat, the only squat exercise admissible at hhr. With cns gone, the two resource-cost rows

are $R_{\text{back squat}} = (\text{knee:3}, \text{post:1}, \text{trunk:2})$ and $R_{\text{strict press}} = (\text{shldr:3}, \text{trunk:1})$, sharing only trunk:

$$\widehat{R}_{\text{back squat}} \cdot W \cdot \widehat{R}^{\top}_{\text{strict press}} = \frac{2 \times 1}{\|(3,1,2)\|\ \|(3,1)\|} = \frac{2}{3.742 \times 3.162} = 0.169.$$

Back squat and strict press share no pattern (squat $\neq$ push), so the same-pattern-repeat term is 0. Pattern squat was touched 3 days ago by that same back-squat set, which also satisfies squat's own loaded clock, so (squat, hhr) is not overdue. $\widehat{D}_{\text{squat,hhr}} = 1.2/1.2 = 1.0$ (squat has never had an hhr exposure, so this cell is fully unmet):

$$\begin{aligned}\text{score}(\text{squat}, \text{hhr}) = &\underbrace{1.0}_{\widehat{D}} + \underbrace{0.35 \times 0.833}_{0.292} + \underbrace{0.25 \times \tfrac{21}{21}}_{0.250} + \underbrace{0.10 \times \tfrac{1.8}{3.0}}_{0.060} \\ &+ 0 - \underbrace{0.25 \times 0.169}_{0.042} - 0 \\ &= 1.559.\end{aligned}$$

Comfortably positive, and this is the direct payoff of the two changes in Section 8's reversal: under the raw, pre-`cns`-removal formula this same candidate scored -0.398 — a 2-point overlap penalty (driven mostly by cns:$3 \times 2 = 6$ of the raw dot product's 8) exceeded every positive term combined. With `cns` gone and the remaining overlap unit-normalized, the penalty drops to 0.042: back squat and strict press turn out to share very little resource cost once the artificially-universal `cns` entry is removed, and the residual overlap costs a proportionate amount rather than dominating the score.

This is also where the hard rule bites: (push, hhr), with a normalized deficit of $0.8/1.2 = 0.667$, would ordinarily be a live candidate in the workout pool — but because push was already loaded today (by strict press, at heavy), that cell is *dropped from the candidate list entirely*, not merely penalized. It cannot appear no matter how far behind it is, for the rest of this session.

12.6 The `recommend()` verdict on this state

Using the same deficit table, at the original state (before either pick above):

$$\begin{aligned}\rho_{\text{light}} &= 100 \cdot \frac{7.1 + 8.1 + 8.1}{8.1 \times 3} = 95.9\%, \\ \rho_{\text{heavy}} &= 100 \cdot \frac{2.8 + 2.0 + 2.0 + 2.0}{3.0 \times 4} = 73.3\%, \\ \rho_{\text{hhr}} &= 100 \cdot \frac{0.8 + 1.2 + 1.2 + 1.2 + 1.0}{1.2 \times 5} = 90.0\%.\end{aligned}$$

All three are positive (nothing is over-covered), so the sign guard from Section 11 does not apply. Light ranks first, hhr second, margin $95.9 - 90.0 = 5.9$ — under the leaning threshold of 10, so the verdict is `no strong signal` with no style named, even though light's raw deficit sum (23.3) is nearly double heavy's (8.8). That gap is exactly why the comparison is relative to each style's own target rather than on raw deficits: light's target is also nearly triple heavy's, and once that is accounted for, the three styles are much closer to equally served than the raw sums suggest.

Summary: every item flagged above

constant	value used	status
`WINDOW_DAYS`	42	GUIDE.md value, not second-guessed
`TARGET_RATE["hhr"]`	0.20	midpoint of GUIDE.md's 0.15–0.25 range
`HHR_CREDIT`	0.5	GUIDE.md value; "lower if priority shifts" is your call
`OVERDUE_DAYS`	7 / 14 / 24	GUIDE.md values, stated as goals-as-constraints
`GAMMA_BEHIND` / `GAMMA_OVER`	0.35 / 0.10	tuned/near-noise per GUIDE.md's own classification (corrected — previously mislabeled "trust"); asymmetry is the point, magnitudes aren't
`LAMBDA_CELL`	0.25 (was 0.10)	rescaled for normalized [0,1] deficit; still a placeholder
`LAMBDA_MOVE`	*removed*	simplification; selection is unaffected, ablation found it near-noise
`LAMBDA_UNI`	0.10 (was 0.30)	rescaled for normalized [0,1] deficit; still a placeholder
`UNILATERAL_SHORTFALL_CAP`	3.0	unspecified in GUIDE.md; placeholder (defines what gets normalized, unchanged)
resource matrix W	identity (8×8)	unspecified in GUIDE.md; conservative default; `cns` row/col removed, `hang` row/col added
`RESOURCE_OVERLAP_COEF`	0.25	tuned/near-noise per GUIDE.md; same value, new meaning — weights bounded cosine similarity (Section 8's reversal), not raw magnitude
`SAME_PATTERN_REPEAT_COEF`	1.1	provenance unclear — not in GUIDE.md's near-noise list nor its trust-these table; still a raw count, not normalized by session size
`OVERDUE_BONUS`	1000.0	sentinel, not derived from other terms; even more headroom now that other terms are normalized
`REP_SCHEMES`	generic list	not specified in GUIDE.md; placeholder menu
band thresholds	25 / 10	GUIDE.md values
E/R dictionary rows	authored draft beyond the 3 GUIDE.md examples — separate review	

None of the placeholder rows above have been exercised against real training data. Per `docs/training/GUIDE.md`'s own build order, that is the next step, not more constant-tweaking in the abstract. The resource-overlap coefficient's disproportionate power (an earlier version of this document showed it flipping a candidate's score negative) has since been resolved by dropping `cns` and normalizing interference to cosine similarity, both described in Section 8 and demonstrated in Section 12. The same-pattern-repeat coefficient's analogous issue — a raw count rather than a normalized fraction — has not.